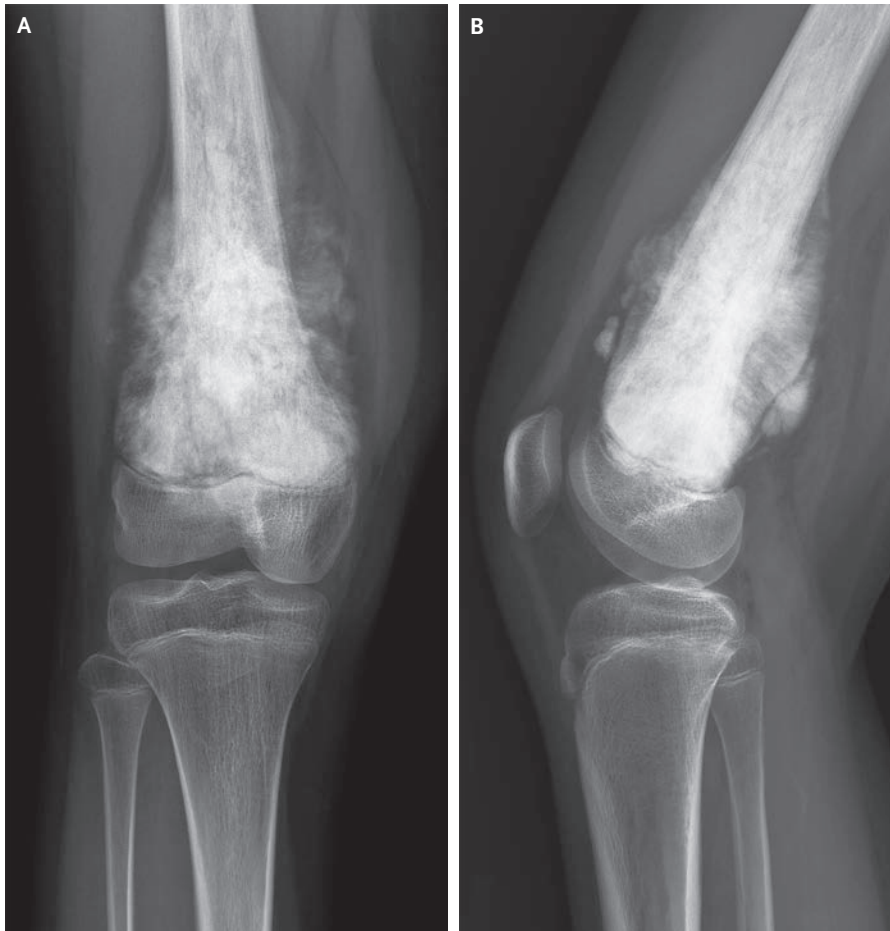


IMAGES IN CLINICAL MEDICINE

Osteosarcoma



AN 11-YEAR-OLD GIRL PRESENTED TO THE ORTHOPEDIC CLINIC WITH A 1-month history of right thigh pain that worsened at night. She had no systemic symptoms. Physical examination was notable for soft-tissue swelling over the distal thigh and limited range of motion of the knee on the right side. Radiographs of the knee showed a metaphyseal osteoblastic lesion with ill-defined margins and cortical destruction in the distal femur (Panel A, anteroposterior view; Panel B, lateral view). The mass had a sunburst appearance, a finding that is seen in aggressive bone lesions and that results from the formation of divergent bone spicules as the periosteal reaction to the rapid growth of the tumor. On the basis of subsequent magnetic resonance imaging of the knee and a bone biopsy, a diagnosis of high-grade osteosarcoma was made. Osteosarcoma is the most common primary malignant bone tumor in children. It typically manifests in the second decade of life owing to the rapid bone growth during that stage of development. No distant sites of metastasis were identified on further imaging. Treatment with neoadjuvant chemotherapy, wide excision, reconstruction, and adjuvant chemotherapy was given. After more than 10 years of follow-up, the patient remained cancer-free with independent ambulation.

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